Project B: AI Instructor Toolkit Dashboard (POC)

1) Background and purpose

McGraw Hill wants a POC that demonstrates instructor-facing value quickly, grounded in adopted content and aligned with guardrails and quality expectations. This project builds the **Instructor Toolkit Dashboard**—a workflow-oriented product that uses Project A's engine to create high-impact instructor outputs.

2) Objectives (POC outcomes)

By the end of the performance period, Project B will deliver:

1. A working **Instructor Toolkit Dashboard** that supports **two flagship instructor workflows** (POC scope) powered by Project A's engine:
    - **Lecture/Slide Scaffolding Tool** (instructional design aligned)
    - **Assessment Transformation Tool** (MCQ → open-ended + rubric aligned to objectives)
2. An instructor-in-the-loop experience: preview → edit → approve → export/publish.
3. A structured **pilot test with a select group of instructors** (recruited/organized by sponsor), capturing feedback and iteratively refining the toolkit.

3) In-scope scope (what will be built)

A. Dashboard experience (POC UI + workflow)

- Web-based dashboard (simple, POC-grade UI; **no WCAG requirement**) that enables:
    - select template/workflow,
    - input parameters (learning objectives, audience, time, tone),
    - choose content scope (chapter/sections),
    - optionally enable/disable external resources (default off for POC unless sponsor approves),
    - preview results with citations/provenance,
    - edit outputs in-line,
    - export artifacts (formats below).

B. Workflow 1: Lecture/Slide Scaffolding Tool (Instructional Design best practices)

This tool must follow **instructional design best practices** at the acceptance-criteria level. The POC will adopt a primary framework (recommended: **Backward Design/Ubd**) and optionally map learning objective verbs to **Bloom's taxonomy**.

Expected outputs (POC):

- lesson/lecture outline aligned to objectives,
- key concepts + misconceptions,
- checks-for-understanding aligned to objectives,
- suggested in-class activity ideas (time-boxed),
- optional slide-by-slide outline and speaker-note draft scaffolding (not "perfect deck in one shot"),
- citations back to OER sections where relevant.

Deliverable artifact: **Instructional Design Template Pack** (short guide + examples) used to shape prompts, UI inputs, and output checks.

C. Workflow 2: Assessment Transformation Tool (Assessment design best practices)

- Transform instructor-provided MCQs (or question sets) into:
  - open-ended prompts aligned to objectives,
  - grading rubric (criteria + performance levels),
  - rationale/expected response outline,
  - common misconceptions/partial credit guidance (optional if time allows).
- Provide **assessment quality checklist** (item clarity, alignment, cognitive level, ambiguity avoidance) used in pilot evaluation.

D. Export/publish (POC-level)

- Export to:
  - structured text/markdown,
  - doc/PDF-friendly format,
  - optionally PPTX-compatible outline format (POC-level; not full design automation),
  - package artifacts for instructor review.
- "Publish" capability (POC-level): mark an artifact/tool output as "approved" for a course context (no LMS integration).

4) Out of scope (explicit exclusions)

- Building all four original tool categories (scope is **two flagship tools only**).
- Full slide deck design with polished visuals/templates comparable to production courseware.
- Automated grading of student submissions at scale.
- Full analytics suite beyond basic usage metrics needed for the pilot.
- Platform integrations (Connect/SimNet/LMS) beyond standalone export artifacts.
- Accessibility compliance standards (WCAG) not required.

5) Assumptions and constraints

- **Dependency:** Project B depends on Project A for:
  - engine API availability,
  - content retrieval and citations,
  - guardrail behaviors,
  - stable provider/quota path for demos/pilot.
- **Content constraint:** POC uses **OER** content selected in Project A.
- **Pilot constraint:** Sponsor will recruit/coordinate a select instructor cohort for pilot testing.
- **Model/provider:**
  - dashboard development may proceed against mock engine endpoints early,

    - pilot and formal sponsor demos must run on stable, project-owned/sponsor-provisioned credentials with defined quotas.

6) Success metrics and acceptance criteria (POC "Definition of Done")

Project B will be accepted when:

- **Workflow completion:** Instructors can complete both flagship workflows end-to-end (generate → edit → approve → export).
- **Instructional design alignment:** Lecture scaffolding outputs consistently reflect the chosen framework (e.g., objectives → aligned activities/checks).
- **Assessment quality:** Transformed assessments include objective alignment + rubric and meet checklist standards in pilot sampling.
- **Provenance:** Outputs show citations/provenance back to OER sources (where applicable).
- **Pilot executed:** Select instructor pilot completed; results summarized; refinements implemented.
- **Demo readiness:** System supports four formal sponsor presentations and final demo.

7) Quality & evaluation plan (includes instructor pilot)

A. Pilot design (select instructor cohort)

- Sponsor recruits a small cohort (e.g., 5–12 instructors) suitable for POC feedback.
- Team designs a lightweight pilot protocol:
    - task-based scripts (e.g., "create a lecture outline for chapter X," "convert quiz to open-ended + rubric"),
    - short feedback instrument (usefulness, clarity, edit burden, trust/citations, likelihood of reuse),
    - capture "time to acceptable output" and common friction points.

B. Iteration plan tied to sponsor cadence

- Pilot run window planned between Presentation #2 and #3 (May → June) so refinements can be incorporated before final demo.
- Pilot results used to prioritize fixes:
    - prompt/template adjustments,
    - UI improvements,
    - output formatting and export improvements,
    - guardrail/grounding clarifications (in coordination with Project A).

C. Equivalent validation for the dashboard

- Usability checks (task completion, error rates, friction points)
- Output quality sampling using instructional design + assessment checklists
- Reliability checks during demo/pilot (latency, failure handling)

8) Governance, roles, and cadence

Cadence:

- **Bi-weekly sponsor meetings** with demo of current state, review of pilot progress, and prioritization.
- Formal presentations at the four specified weeks (see schedule).

9) Schedule and milestones (aligned with bi-weekly sponsor cadence)

Phase 0 — SOW Revision (Pre-execution)

- **1/26 – 2/13:** Revised SOW finalized (deliverable due **2/13**).

Phase 1 — Execution (students begin week of 2/16; through 8/12)

- **Sprint 1:** Dashboard skeleton + mock engine integration + workflow wireframes; define instructional design framework and draft template pack
- **Sprint 2:** Implement Workflow 1 v0 and Workflow 2 v0 against mock engine; export basics
- **Sprint 3 (Presentation week of 3/23): Presentation #1** — Vertical slice demo (one workflow end-to-end) + integration plan; switch to live Engine v0 when available
- **Sprints 4–6 (to Presentation week of 5/4):** Both workflows fully functional, edit/approve UX, export improvements, pilot protocol ready
- **Sprint 6 (Presentation week of 5/4): Presentation #2** — Full toolkit demo + pilot plan sign-off + readiness check (provider/quota stable)
- **Sprints 7–9 (to Presentation week of 6/15):** Conduct instructor pilot + analyze feedback + implement refinements
- **Sprint 9 (Presentation week of 6/15): Presentation #3** — Pilot results + refined toolkit demo
- **Sprints 10–13 (to Presentation week of 8/10; end 8/12):** Hardening, documentation, final UX polish, final demo script
- **Sprint 13 (Presentation week of 8/10): Presentation #4** — Final demo + handoff package; **8/12** complete

10) Project B deliverables (summary)

- Revised SOW (Phase 0) — due 2/13
- Dashboard MVP (POC UI) with edit/approve/export
- Lecture/Slide Scaffolding workflow (ID-aligned) + **Instructional Design Template Pack**
- Assessment Transformation workflow + **Assessment Quality Checklist**
- Pilot protocol + instruments + recruitment assumptions
- Pilot execution + results summary + refinement backlog

- Final demo script + final toolkit handoff package

11) Explicit dependency on Project A (and how it will be managed)

- Project B will build against **mock engine endpoints** until Engine v0 is delivered.
- Project A must provide:
  - API contract early,
  - Engine v0 by week of **3/23**,
  - stable provider/quota path by week of **3/23**.
- Integration will be verified through:
  - contract tests,
  - end-to-end workflow demos at sponsor meetings,
  - a shared change log for any breaking changes.